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Solution curve

Definition

The graph of a solution of a differential equation is called a **solution curve**.

More generally, a curve C is said to be an **integral curve** of a differential equation if every function $y = y(x)$ whose graph is a segment of C is a solution of the differential equation.

Thus, any solution curve of a differential equation is an integral curve (**Why?**)

But an integral curve need not be a solution curve.

Example

If R is any positive constant, then the circle

$$x^2 + y^2 = R^2 \tag{1}$$

is an integral curve of the differential equation

$$y' = -\frac{x}{y}. \tag{2}$$

Solution

One one hand, we note that the only functions whose graphs are segment of (1) are

$$y_1 = \sqrt{R^2 - x^2} \quad \text{and} \quad y_2 = -\sqrt{R^2 - x^2}.$$

On the other hand, we can easily verify that these functions both satisfy (2) on the open interval $(-R, R)$.

Indeed, for $y_1 = \sqrt{R^2 - x^2}$ we have

$$\begin{aligned} y_1' &= \frac{1}{2\sqrt{R^2 - x^2}} \cdot (R^2 - x^2)' = \frac{-2x}{2\sqrt{R^2 - x^2}} \\ &= -\frac{x}{\sqrt{R^2 - x^2}} = -\frac{x}{y_1}. \end{aligned}$$

For $y_2 = -\sqrt{R^2 - x^2}$, it is similar.

However, (1) is not a solution curve of (2). **Why?**

The reason is it is not the graph of a function.

Example

Verify that

$$y = \frac{x^2}{3} + \frac{1}{x} \tag{3}$$

is a solution of

$$xy' + y = x^2 \tag{4}$$

on $(0, \infty)$ and $(-\infty, 0)$.

Solution

Substituting (3) and $y' = \frac{2x}{3} - \frac{1}{x^2}$ into (4) yields

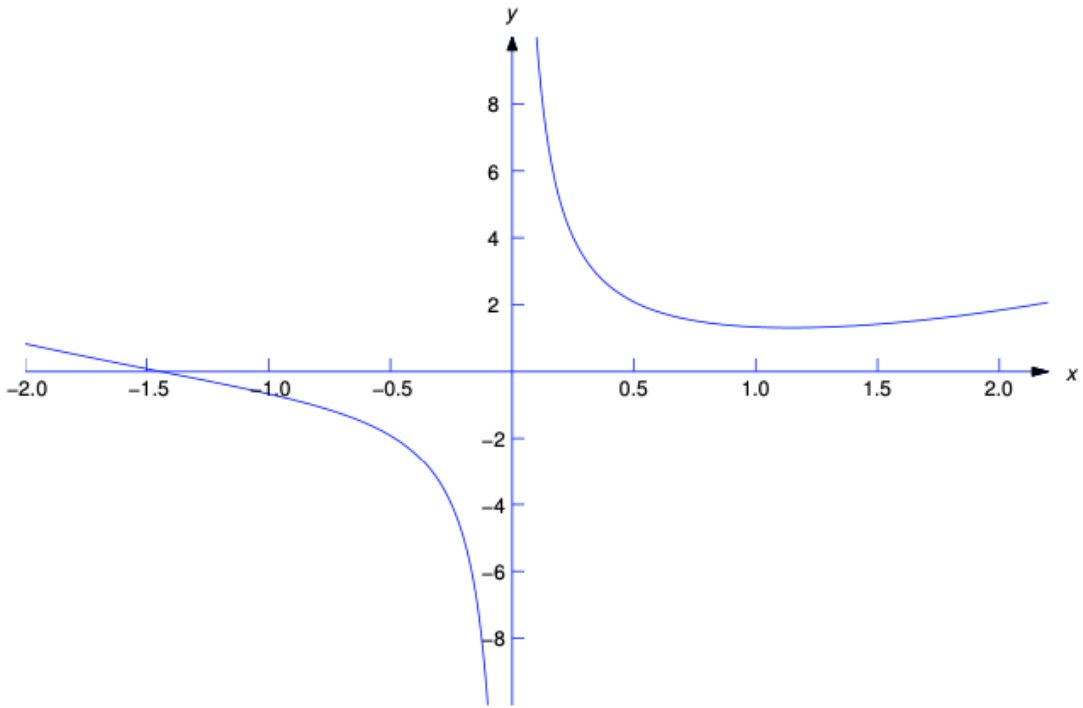
$$xy' + y = x\left(\frac{2x}{3} - \frac{1}{x^2}\right) + \left(\frac{x^2}{3} + \frac{1}{x}\right) = x^2$$

for all $x \neq 0$.

Thus y is a solution of (4) on $(-\infty, 0)$ and $(0, \infty)$.

However, y is not a solution of the differential equation on any open interval that contains $x = 0$, because y is not defined at $x = 0$.

Figure below shows the graph of (3).



The part of the graph of (3) on $(0, \infty)$ is a solution curve of (4), and the same for the part of the graph on $(-\infty, 0)$.

Initial-Value Problem

The unique specification of an applied problem requires more than just a differential equation.

Of particular interest is the case of the initial-value problem (briefly, IVP) defined for an nth-order differential equation.

Let's consider the following example.

Example

Find all solutions of the equation

$$y^{(n)} = e^{2x}.$$

Solution

Integrating the given equation yields

$$y^{(n-1)} = \frac{e^{2x}}{2} + k_1,$$

where k_1 is an arbitrary constant.

If $n \geq 2$, integrating again yields

$$y^{(n-2)} = \frac{e^{2x}}{4} + k_1 x + k_2,$$

where k_2 is another arbitrary constant.

If $n \geq 3$, repeatedly integrating yields

$$y = \frac{e^{2x}}{2^n} + k_1 \frac{x^{n-1}}{(n-1)!} + k_2 \frac{x^{n-2}}{(n-2)!} + \cdots + k_n,$$

where $k_1, k_2, \ldots, k_n$ are arbitrary constants.

Since the constants $k_1, k_2, \ldots, k_n$ are arbitrary, so are the constants

$$\frac{k_1}{(n-1)!}, \frac{k_2}{(n-2)!}, \cdots, k_n.$$

Thus all solutions of the given equation can be written as

$$y = \frac{e^{2x}}{2^n} + C_1 + C_2 x + \cdots + C_n x^{n-1},$$

where $C_1, C_2, \ldots, C_n$ are arbitrary constants.

The above example shows that the differential equation $y^{(n)} = e^{2x}$ has, as expected, an infinite family of solution that depend upon the n arbitrary constants $C_1, C_2, \ldots, C_n$.

In the absence of additional conditions, there is no reason to prefer one solution of a differential equation over other.

However, we'll often be interested in finding a solution of a differential equation that satisfies one or more specific conditions.

The next examples illustrate this.

Example

Solve the equation

$$y' = x^3 \tag{5}$$

so that

$$y(1) = 2. \tag{6}$$

Solution

This is a calculus I problem: integrating both sides of (5), we have

$$y = \int x^3 \, dx = \frac{x^4}{4} + C,$$

where C is an arbitrary constant.

We now impose the auxiliary condition (6).

Setting $x = 1$ and $y = 2$ in the formula above, we get

$$y(1) = 2 \iff 2 = \frac{1}{4} + C \iff C = \frac{7}{4},$$

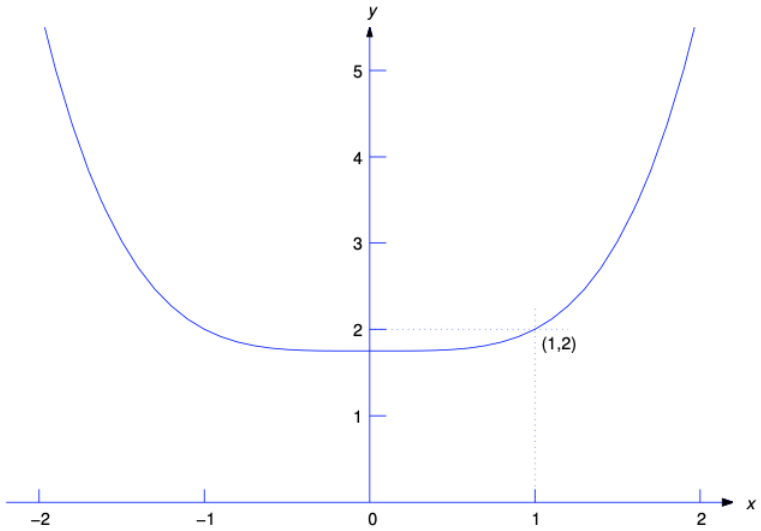
that is $y(x) = \dfrac{x^4 + 7}{4}$.

Hence,

$$y(x) = \frac{x^4 + 7}{4} \tag{7}$$

is the unique solution to the given equation.

The following figure shows the graph of this solution.



Note that imposing the condition $y(1) = 2$ is equivalent to requiring the graph of y to pass through the point $(1, 2)$.

Since the function (7) is defined for all x, the interval of validity of this solution is $(-\infty, \infty)$.

Example

Consider again the differential equation

$$xy' + y = x^2 \tag{8}$$

Solution

We already verified that

$$y = \frac{x^2}{3} + \frac{1}{x} \tag{9}$$

is a solution of (8) on $(0, \infty)$ and on $(-\infty, 0)$.

By evaluating (9) at $x = \pm 1$, we can see that (9) is a solution of each of the following:

(a) equation (8) with the condition $y(1) = \dfrac{4}{3}$

(b) equation (8) with the condition $y(-1) = -\dfrac{2}{3}$.

The interval of validity of (9) as a solution of (a) is $(0, \infty)$, since this is the largest interval that contains $x = 1$ on which (9) is defined.

Similarly, the interval of validity of (9) as a solution of (b) is $(-\infty, 0)$, since this is the largest interval that contains $x = -1$ on which (9) is defined.

Note. The term initial-value problem originated in problems of motion where the independent variable is t (representing elapsed time), and the initial conditions are the position and velocity of an object at the initial (starting) time of an experiment.

We consider the following example called "free fall under constant gravity".

Example

An object falls under the influence of gravity near Earth's surface, where it can be assumed that the magnitude of the acceleration due to gravity is a constant g.

(a) Construct a mathematical model for the motion of the object in the form of an initial-value problem, assuming that the altitude and velocity of the object at time $t = 0$ are known. Assume that gravity is the only force acting on the object.

(b) Solve the initial value problem derived in (a) to obtain the altitude as a function of time.

Solution

(a) Denote by $y(t)$ the altitude of the object at time t. Since the acceleration of the object has constant magnitude g and is in downward (negative) direction, y satisfies the second order equation

$$y'' = -g.$$

We denote y_0 and v_0 the altitude and velocity when $t = 0$. Then y is a solution of the IVP

$$y'' = -g, \quad y(0) = y_0, \quad y'(0) = v_0. \tag{10}$$

(b) Integrating (10) twice, we get

$$y' = -gt + C_1 \implies y = -\frac{gt^2}{2} + C_1 t + C_2.$$

By initial conditions, we have

$$y(0) = C_2 = y_0 \quad \text{and} \quad y'(0) = C_1 = v_0.$$

Hence, the solution of the IVP (10) is

$$y(t) = -\frac{gt^2}{2} + v_0 t + y_0.$$

Now we introduce the following.

Definition

An nth-order differential equation together with n auxiliary conditions of the form

$$\begin{aligned} y(x_0) &= y_0 \\ y'(x_0) &= y_1 \\ &\dots \\ y^{(n-1)}(x_0) &= y_{n-1}, \end{aligned}$$

where $y_0, y_1, \dots, y_{n-1}$ are constants, is called an **initial-value problem**.

We have the following important theorem.

Theorem

Let $a_1, a_2, \ldots, a_n, F$ be functions that are **continuous** *on an interval I. Then, for any $x_0 \in I$, the initial-value problem*

$$y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y = F(x),$$

$$y(x_0) = y_0, y'(x_0) = y_1, \ldots, y^{(n-1)}(x_0) = y_{n-1}$$

has a **unique** *solution on I.*

Theorem about IVP for first-order differential equation

Let's consider the following IVP

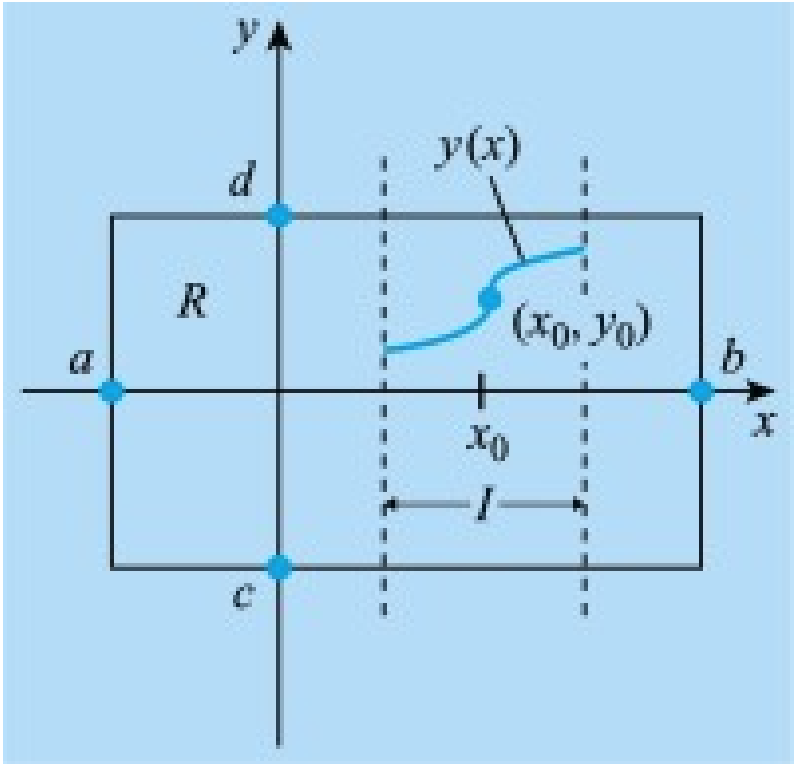
$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \tag{11}$$

If $f(x, y)$ is continuous in the closed rectangle

$$R = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$$

and $\dfrac{\partial f}{\partial y}$ is also continuous in R, then for any interior point (x_0, y_0) in R, there exists an interval I containing x_0 such that the IVP (11) has a **unique** solution on I.

The following figure gives an idea of what such a region R and interval I may look like.



Some important comments about this fundamental theorem.

1. If the conditions in the theorem are satisfied, then solution curves for the IVP can never intersect. (**Why?**)
2. If $f(x, y)$ and $\dfrac{\partial f}{\partial y}$ are continuous for all values of x and y, the result does not say that the unique solution must be valid for all values of x and y.
3. The continuity of $f(x, y)$ and $\dfrac{\partial f}{\partial y}$ are sufficient for the existence of solutions, but they may not be necessary to guarantee their existence. $\Longrightarrow$ This means that we may have solutions even if the continuity condition is not satisfied.
4. This is an existence theorem, which means that if the right conditions are satisfied, we can find a solution, but we are not told how to find it. $\Longrightarrow$ In particular, we may not be able to describe the interval I without actually solving the differential equation.

Remark. If $f(x, y)$ satisfies the hypotheses of the existence and uniqueness theorem in a region R of the xy-plane, then the solution curves to a differential equation

$$\frac{dy}{dx} = f(x, y)$$

cannot intersect in R.

In other words, from a geometric point of view, if $f(x, y)$ satisfies the hypotheses of the IVP theorem in a region R of the xy-plane, then throughout that region the solution curves of the differential equation $\frac{dy}{dx} = f(x, y)$ cannot intersect.

• Equilibrium solutions

Definition

Any solution to the differential equation $\frac{dy}{dx} = f(x, y)$ of the form $y(x) = y_0$, where y_0 is a constant, is called an equilibrium solution to this equation.

From this definition, it follows that

- the corresponding solution curve is a line parallel to the x-axis.
- equilibrium solutions are given by any constant values of y, for which $f(x, y) = 0$.

We have the following note.

Note. No solution to the differential equation

$$\frac{dy}{dx} = f(x, y)$$

can intersect with equilibrium solution of the differential equation.

Indeed, an equilibrium solution is a solution, and two solution curves to the differential equation $\frac{dy}{dx} = f(x, y)$ do not intersect as already explained above.

Consequently, equilibrium solutions serve to divide the xy-plane into different regions.

Question.

Does every differential equation $\dfrac{dy}{dx} = f(x, y)$ have at least one equilibrium solution?

Answer.

NO (!)

A counterexample?

The differential equation

$$\frac{dy}{dx} = y - x.$$

has no equilibrium solution.

We provide one interesting application of IVP - **a self-study (!)**

Example

The general solution to the differential equation

$$y'' + \omega^2 y = 0, \ -\infty < x < \infty \tag{5}$$

where ω is a nonzero constant, is

$$y(x) = C_1 \cos \omega x + C_2 \sin \omega x, \tag{6}$$

where C_1, C_2 are arbitrary constants.

Solution

According to the definition of the general solution, we have to show the following:

1. $y(x) = C_1 \cos \omega x + C_2 \sin \omega x$ is a solution to the differential equation (5) on $(-\infty, \infty)$.
2. *every* solution to (5) is of the form (6).

The first claim is easy (!) to verify.

For the second claim, take an arbitrary solution $y = f(x)$ of (5). Let $f(0) = A,\ f'(0) = B$. By the theorem above, the initial-value problem

$$\begin{cases} y'' + \omega^2 y = 0 \\ y(0) = A,\ y'(0) = B \end{cases} \tag{7}$$

has a unique solution.

Notice that $y = f(x)$ satisfies this problem, and so $y = f(x)$ is precisely that unique solution to the initial-value problem (7).

Consider the function

$$\hat{y}(x) = A\cos\omega x + \frac{B}{\omega}\sin\omega x. \tag{8}$$

This function is of the form (6), and so, by the first claim, is a solution to equation (5).

We notice that evaluations of the function (8) and its derivative at $x = 0$ are as follows

$$\hat{y}(0) = f(0) = A,\ \hat{y}'(0) = f'(0) = B,$$

which shows that $\hat{y}(x)$ is also a solution to the initial-value problem (7).

So these two solutions must coincide. Therefore,

$$\begin{aligned} f(x) &= A\cos\omega x + \frac{B}{\omega}\sin\omega x \\ &= f(0)\cos\omega x + \frac{f'(0)}{\omega}\sin\omega x \\ &= C_1\cos\omega x + C_2\sin\omega x. \end{aligned}$$

Since $f(x)$ is an arbitrary solution to the differential equation (5), we can conclude that every solution to (5) is of the form (6), that is

$$y(x) = C_1\cos\omega x + C_2\sin\omega x$$

and therefore this is the general solution on $(-\infty, \infty)$.

□

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Separable Differential Equations

We provide some **solution techniques** for determining the exact solution to certain types of differential equations.

The simplest differential equations for which a solution technique can be obtained are the so-called separable equations.

Definition

A first-order differential equation is called **separable** if it can be written in the form

$$p(y)\frac{dy}{dx} = q(x) \tag{1}$$

(that is, if we can separate $p(y)\,dy$ and $q(x)\,dx$).

The solution technique for a separable differential equation is given in the following result.

Theorem

If $p(y)$ and $q(x)$ are **continuous**, *then Equation (1) has the* **general solution**

$$\int p(y)\,dy = \int q(x)\,dx + C, \tag{2}$$

where C is an arbitrary constant.

Exercise

Prove this theorem (!)

Remark

In differential form Equation (1), as noted above, can be written as

$$p(y)\,dy = q(x)\,dx,$$

and the general solution (2) is obtained by integrating the left-hand side w.r.t. y and the right-hand side w.r.t. x.

This is the **general procedure** for solving separable equations.

Example

Solve $(1 + y^2)\dfrac{dy}{dx} = x \cos x$.

Solution

This differential equation is separable: $(1 + y^2)\, dy = x \cos x\, dx$.

Integrating both sides of the differential equation yields

$$\int (1 + y^2)\, dy = \int x \cos x\, dx + C.$$

First to evaluate the integral on the right-hand side, we use integration by parts

$$\int u\, dv = uv - \int v\, du.$$

Just for $\int x \cos x \, dx$, we put

$$\begin{cases} u = x \\ dv = \cos x \, dx \end{cases} \Longrightarrow \begin{cases} du = dx \\ v = \sin x, \end{cases}$$

and so

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx = x \sin x + \cos x + C.$$

Next, the anti-derivative of the expression under the integral in the left-hand side $\int (1 + y^2) \, dy$ is

$$y + \frac{y^3}{3}.$$

Therefore, we have

$$y + \frac{y^3}{3} = x \sin x + \cos x + C,$$

or equivalently

$$y^3 + 3y = 3(x \sin x + \cos x) + C_1,$$

where $C_1 = 3C$.

As often happens with separable differential equations, the solution is given in implicit form.

□

Remark

In general, the differential equation $\frac{dy}{dx} = f(x)g(y)$ is separable, since it can be written as

$$\frac{1}{g(y)}\frac{dy}{dx} = f(x),$$

which is of the form of Equation (1) with $p(y) = \frac{1}{g(y)}$.

However, it is important to note that in writing the given differential equation in this way, we have assumed that $g(y) \neq 0$. Thus the general solution to the resulting differential equation may not include solutions of the original equation corresponding to any values of y for which $g(y) = 0$.

We illustrate with an example.

Example

Find all solutions to

$$y' = -2y^2x. \tag{3}$$

Solution

Separating the variables yields

$$\frac{dy}{y^2} = -2x\ dx. \tag{4}$$

Integrating both sides, we obtain

$$-\frac{1}{y} = -x^2 + C$$

so that

$$y(x) = \frac{1}{x^2 - C}.$$

This is the general solution to Equation (4), but it is not the general solution to Equation (3), since there is no value of the constant C for which $y(x) = 0$, whereas $y(x) = 0$ is a solution to Equation (3).

Thus the solutions to Equation (3) are

$$y(x) = \frac{1}{x^2 - C} \quad \text{and} \quad y(x) = 0,$$

where C is an arbitrary constant.